

2024



Amanote Optimizes Note-Taking App for Learners, Handling Millions of Queries on AWS

[Amanote](#) can onboard new large university customers for its note-taking app in less than 2 minutes, scale to accommodate thousands of new users in a few hours, and handle up to 3 million queries per customer per month. Its app is integrated into learning management systems (LMS) used in higher-education facilities and large enterprises. Amanote needed to ensure that its app could be easily integrated as an LMS plugin, with customization kept to a minimum to assist sales. It built a fully serverless cloud setup, using AWS Lambda to handle queries and deliver multiple app features. Amanote has optimized its costs and can easily deliver its products to customers worldwide with minimal disruption.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



20K

users added each month

1,000

of new customers



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Up to 3 Million

2 Minutes



Overview

Belgium-based educational technology (EdTech) business Amanote created a note-taking app that's used by students to annotate course materials. With plans to grow its customer base in higher-education facilities, it wanted to make sure its infrastructure could grow with customer demand. From the start, Amanote decided to build on Amazon Web Services (AWS) to benefit from its comprehensive set of serverless services, helping it to scale rapidly and support growth.

Using AWS, Amanote can onboard a new university customer in 2 minutes and accommodate thousands of users in just a few hours. It has achieved this by building the app's backend and multiple features on AWS Lambda so it can run code without thinking about servers or clusters.

With pay-as-you-go services from AWS, the company can focus on optimizing service use to keep costs low and also minimize the number of API calls made. This has not only minimized costs, but also improved quality of service and app performance for end users.



Opportunity | Benefiting from AWS Startups to Support App Development and Product Versatility

Amanote is helping transform learning in both higher-education and staff training scenarios. Its app makes it easy to add notes and comments to coursework. This includes slide-, PDF- and video-based educational content that is created for learning management systems (LMS). Amanote helps educators and instructors to deliver more interactive lessons, add oral explanations, and gain feedback to help improve course content. For learners, the app boosts learning productivity by making note taking more effective. The app helps to generate color-coded note summaries, and supports note sharing with other learners—both of which help students to enhance their study and revision practice.

The inspiration for Amanote came from the direct experience of founder and chief executive officer, Fery, while he was studying for a master's degree in computer science. He saw a need to digitize and make the learning process more efficient and started to develop a solution. Because of the complex technology

requirements of many universities' legacy IT systems, Fery decided to offer Amanote as a software plugin, so it could be installed easily to avoid the need for lengthy software implementations. The company was launched in 2016, with the new app built entirely on AWS. "We chose AWS because of its superior technology and its broad set of serverless services," he says. "This gave us all the components we needed to develop a fully serverless environment—with zero compromise."

When it first launched, Amanote offered a note-taking app that students could download and use independently. In 2020, the company began to work on developing its app so it could be integrated with LMS providers' software, and the company could offer a comprehensive note-taking solution directly to universities and higher-education institutions—to further enhance the student learning experience and offer more advanced features. For example, by supporting continuous syncing of course content with a student's notes, even if the course content is updated.

To support this move and aid app development, Amanote applied for [AWS Activate](#) funding, which provides AWS credits at no cost to help provide startups with the resources they need to build, launch, and scale on AWS. The program also provided access to education events to help the company find potential new customers and network with peers. Amanote also received support from AWS solution architects to assist with architecture design. "We had access to AWS support, which we didn't expect as a startup," says Fery. "With the help of AWS, we've developed an affordable solution that's versatile and easy for students to use."

Solution | Using AWS Lambda to Handle Millions of Queries and Accommodate Thousands of New Users

Amanote can now scale automatically to accommodate the 20,000 new users it adds each month. It can also set the system to automatically power down during university and college vacations, when usage is low. AWS Lambda sits at the core of the app, handling the 1–3 million queries per customer, per month, that are generated. These queries support several app features. They include setting up user accounts, saving individual notes, and collaborating with fellow students by exchanging information about coursework. "We needed to focus on developing new product features, not managing infrastructure," says Fery. "Using AWS serverless services means we can free up time to focus on what matters—developing and growing the business."

All API requests are sent first to [Amazon API Gateway](#), which the company can use to create, manage, and secure APIs at any scale. This helps ensure that all communications are connected to the right AWS Lambda function. From the start, Amanote has used a cost-effective AWS database solution to store document metadata and manage



licensing using [Amazon DynamoDB](#), a serverless, NoSQL, fully managed database with single-digit millisecond performance at any scale. Amanote benefits from high levels of infrastructure security by building on AWS. Machine learning (ML) is employed to quickly expose any threats or unusual user activity, using [Amazon GuardDuty](#), which protects AWS accounts with intelligent threat detection. This assures its customers that the app is secure. The company has achieved low latency connections globally by using [AWS Global Infrastructure](#), which provides a secure, extensive, and reliable global cloud infrastructure for all applications. This means that it can deliver its product to customers worldwide quickly and securely, without making any system changes.

Many of the company's customers report noticeable results in using the Amanote app, including a key university client in Modena, Italy. "Because of Amanote's ease of implementation, we've been able to get almost immediate results," says Valeria Folloni, Instructional Designer at the University of Modena. "The dedicated budget was minimal and we got tangible results in a very short time, and we're continuing to see improvements." The next stage of development for Amanote is to harness the power of artificial intelligence (AI) capabilities for its app. For internal development work, the company is currently beta testing [Amazon Q](#), a generative AI-powered assistant designed for work that can be tailored to its business. To allow teachers to use their preferred AI large language model, it is also evaluating [Amazon Bedrock](#), to help build and scale generative AI applications with foundation models. It is also starting to test ML enriched data streams, using [Amazon Kinesis Data Firehose](#), to load near-real-time streams into analytics services. This will help to automatically detect when students are struggling with coursework—and provide academic and training staff with insights on where improvements can be made.

Outcome | Growing to Reach 500K Customers in 50 Countries Worldwide

Amanote now has over 500,000 users in 50 countries across four continents, including Asia, Europe, and the US. With a solid app platform, the company is now launching on [AWS Marketplace](#). "We've discovered great benefits using AWS," says Fery. "The documentation is excellent, AWS services are straightforward to use, and we can always find a solution. Everything we need to do, we can do on AWS."

About Company

Belgium-based Amanote offers a note-taking app that integrates with learning management systems. Using annotation that can be classified and shared, the app helps students and business professionals to enhance study and aid revision practice.

AWS Services Used

Amazon Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any application or backend service without provisioning or managing servers.

[Learn more »](#)



AWS API Gateway

Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale.

[Learn more »](#)

Amazon DynamoDB

Amazon DynamoDB is a fully managed, serverless, key-value NoSQL database designed to run high-performance applications at any scale.

[Learn more »](#)

Amazon Aurora

Amazon Aurora provides built-in security, continuous backups, serverless compute, up to 15 read replicas, automated multi-Region replication, and integrations with other AWS services.

[Learn more »](#)

AWS Customer Success Stories

Organizations of all sizes use AWS to increase agility, lower costs, and accelerate innovation in the cloud.

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More education Customer Stories



FHNW Creates Generative AI-Inspired Art Exhibit for...

The University of Applied Sciences and Arts Northwestern (FHNW) is one of Switzerland’s largest universities. In collaboration with interactive museum for analog and digital technology Enter Technikwelt Solothurn, FHNW embarked on a generative artificial intelligence (AI) project using Amazon Web Services (AWS). With managed service Amazon SageMaker, says research associate Claudio Paonessa,



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2024



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2019



Ravensbourne University Case Study

By using AWS Elemental for its video streaming services, Ravensbourne University has been able to reach a higher number of students with higher satisfaction. Ravensbourne is an innovative, industry-focused university located in the heart of London that specializes in the cross-section between creativity and technology. The



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